

Question 2

Consider the simultaneous linear equations

$$mx + 5y = m - 1$$

$$2x + (m - 9)y = 2m + 6$$

- a.** Find the value of m for which there are infinitely many solutions.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

3 marks

- b.** Find the value(s) of m for which there is a unique solution.

1 mark

Total 4 marks

Question 2

Consider the equations $5mx - 5y = 10$ and $3x + (2 - m)y = 6k$ where $m, k \in \mathbb{R}$. Find the values of m and k for which these equations have:

a. a unique solution.

2 marks

b. an infinite number of solutions.

2 marks
